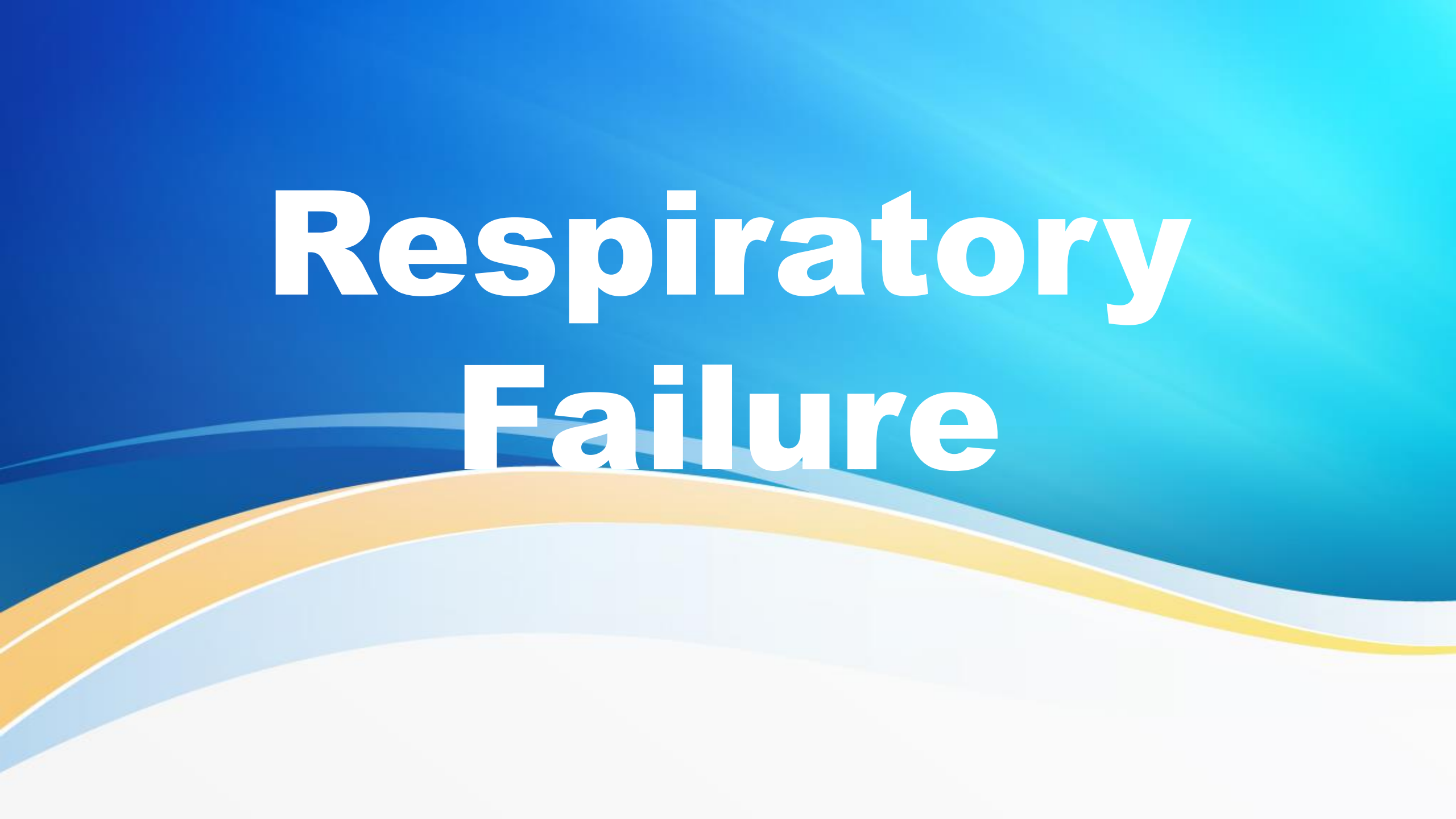


Respiratory Failure

The background of the slide is a blue gradient, transitioning from a darker blue on the left to a lighter blue on the right. At the bottom, there is a decorative wavy line composed of a yellow band and a white band, creating a sense of movement or a horizon line.

Introduction

Respiratory failure is a life-threatening condition in which the respiratory system fails in one or both of its gas-exchange functions:

- Oxygenation (oxygen delivery to the blood)
- Ventilation (removal of carbon dioxide)

It can occur acutely (minutes to hours) or chronically (days to years).

Classification

1. Type I — Hypoxemic Respiratory Failure

A. Based on Gas Exchange Abnormalities

Characterized by:-

- $\text{PaO}_2 < 60 \text{ mmHg}$
- Normal or low PaCO_2

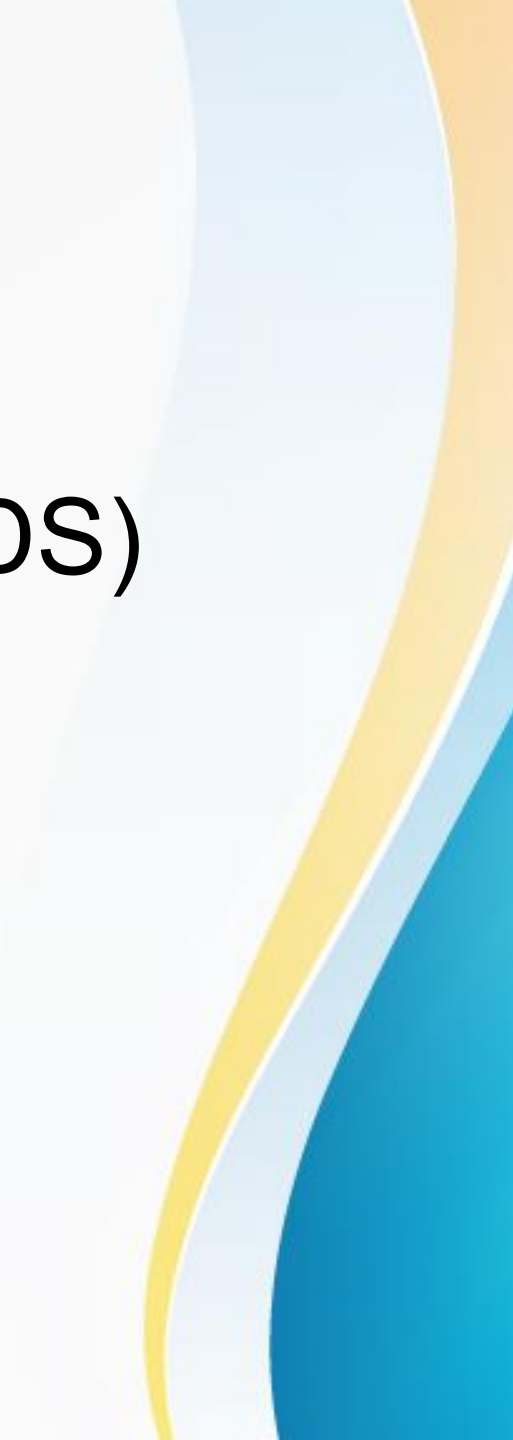
Caused by:-

problems primarily with oxygenation.

Mechanisms:

- V/Q mismatch
- Shunt
- Diffusion impairment
- Low inspired oxygen

Common causes:

- Pneumonia
 - Acute respiratory distress syndrome (ARDS)
 - Pulmonary edema
 - Pulmonary embolism
 - Interstitial lung diseases
- 

2. Type II — Hypercapnic Respiratory Failure

Characterized by:-

- $\text{PaCO}_2 > 45 \text{ mmHg}$
- Often accompanied by acidemia (low pH)

Caused by:-

- hypoventilation (insufficient CO_2 removal).

Mechanisms:

- Decreased respiratory drive
- Airflow obstruction
- Neuromuscular weakness
- Chest wall abnormalities

Common causes:

- COPD exacerbation
- Asthma severe attack
- Drug overdose (opioids, sedatives)
- Neuromuscular disorders (myasthenia gravis, Guillain-Barré)
- Obesity hypoventilation syndrome

3. Type III — Perioperative Respiratory Failure

- Due to atelectasis usually after surgery.
- Occurs commonly after abdominal or thoracic surgery.

4. Type IV — Shock-related Respiratory Failure

- Associated with hypoperfusion, usually in shock.
- Respiratory muscles fatigue due to inadequate oxygen delivery.

Pathophysiology



A. V/Q Mismatch

- Most common mechanism.
- Portions of the lung receive oxygen but not enough blood, or vice versa.
- Improves with supplemental O₂.

B. Shunt

- Blood bypasses ventilated alveoli entirely.
- Does NOT improve significantly with oxygen therapy.

Causes:

- pneumonia,
- ARDS,
- atelectasis

C. Diffusion Limitation

- Thickened alveolar membrane prevents efficient diffusion.
- Seen in interstitial lung diseases.

D. Hypoventilation

- CO₂ accumulation → hypercapnia.
- Causes include sedation, neuromuscular impairment, or chest wall restriction.

4. Clinical Presentation

Symptoms:

- Dyspnea
- Tachypnea
- Use of accessory muscles
- Confusion or agitation (hypercapnia or hypoxemia)
- Cyanosis
- Fatigue and inability to speak in full sentences

Signs of impending respiratory arrest:

- **Altered mental status**
- **Bradypnea or irregular breathing**
- **Silent chest (severe airway obstruction)**
- **Poor respiratory effort**

5. Diagnosis



A. Arterial Blood Gas (ABG)

- Confirms hypoxemia, hypercapnia, and acid–base balance.
- Essential for classifying the type of respiratory failure.

B. Chest Imaging

Chest X-ray or CT to identify pneumonia, edema, pneumothorax, or ARDS.

C. Pulse Oximetry

- Quick assessment of oxygen saturation.

D. Additional Tests

- CBC, electrolytes
- ECG (evaluate for cardiac causes)
- Pulmonary function tests (for chronic cases)

6. Management Principles

Always address:

- 1-Airway
- 2-Breathing
- 3-Circulation

A. General Support

- Oxygen therapy to maintain adequate saturation
- Treat underlying cause (infection, fluid overload, bronchospasm, PE)

B. Non-Invasive Ventilation (NIV)

Useful in:

- 1-COPD exacerbations
- 2-Cardiogenic pulmonary edema
- 3-Select neuromuscular patients

Avoid if:

- 1-Severe acidosis
- 2-Hemodynamic instability
- 3- Inability to protect airw

C. Invasive Mechanical Ventilation

Indicated when:

- NIV fails
- Airway protection is needed
- Severe hypoxemia or hypercapnia with acidemia
- Modes could include volume control, pressure support, etc.

D. Treating Underlying Causes

- Antibiotics for pneumonia
- Bronchodilators + steroids for COPD/asthma
- Diuretics for pulmonary edema
- Anticoagulation for pulmonary embolism
- Reversal of drug overdose (e.g., naloxone)

7. Complications

- Barotrauma (pneumothorax)
 - Ventilator-associated pneumonia
 - Hypoxic organ damage
 - Cardiac arrhythmias
 - Respiratory acidosis/alkalosis
- 

8. Prognosis

Depends on:

- Cause (e.g., COPD vs ARDS)
- Timeliness of intervention
- Presence of multi-organ failure
- Age and comorbidities
- ARDS and sepsis-related respiratory failure have the highest mortality.

Thank You

A decorative graphic on the right side of the slide, consisting of several overlapping, curved, wavy shapes in light blue, yellow, and a darker blue, creating a modern, abstract design.